

Use Case ID	UC – 001		
Use Case Name	Create Trip		
Created by	Ngoc Vu	Last Updated By	Ngoc Vu
Date Created	July 02, 2026	Date Last Updated	July 02, 2026
Primary Actors	Traveler		
Supporting Actors	None		
Goal	Allow travelers to create new trips for future travel planning.		
Trigger	“Create Trip” button is clicked.		
Pre-conditions	Traveler is authenticated. Traveler has permission to create trips.		
Post-conditions	A new trip record is successfully created and displayed in the Trip List.		
Main Flow	Step	Actor Action	System Response
	1	Traveler click “Create” trip.	System display trip information input.
	2	Traveler enters trip information.	System validates required fields in real time.
	3	Traveler click “Create” button.	System validates the input. System creates a new Trip record. System displays a success message. System redirects the traveler to the Trip List.
Alternative Flow		Description	
	AF-01	At step 3, Traveler click “Cancel” button. System discards unsaved changes. System redirects the traveler to the Trip List.	
Exception Flow		Description	
	EF-01	At Step 2, Traveler leaves mandatory fields empty. System highlights missing fields. System displays validation messages.	
Business Rules	Destination is mandatory. Start Date is mandatory. End Date must be later than Start Date. Trip Status defaults to Draft.		
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds. Trip data shall be stored securely. System availability shall be at least 99%.		
Acceptance Criteria	Given a traveler enters valid trip information When clicking Create Then a new trip is created And the trip appears in the Trip List		

	And a success message is displayed And the created trip can be edited afterwards
--	---

Use Case ID	UC – 002		
Use Case Name	Edit Trip		
Created by	Ngoc Vu	Last Updated By	Ngoc Vu
Date Created	July 02, 2026	Date Last Updated	July 02, 2026
Primary Actors	Traveler		
Supporting Actors	None		
Goal	Allow traveler to edit trip so it can remain accurate and up to date.		
Trigger	“ Edit trip ” button is clicked by traveler.		
Pre-conditions	Trip exists. Traveler has edit permission.		
Post-conditions	Updated information about the trip is recorded and displayed in the Trip List.		
Main Flow	Step	Actor Action	System Response
	1	Traveler click “ Edit ” trip.	System display update form.
	2	Traveler fill in updated information.	System validates required fields in real time.
	3	Traveler click “ Save ” button.	System validates the input. System renders updated information. System displays a success message. System redirects the traveler to the Trip List.
Alternative Flow		Description	
	AF-01	At step 3, Traveler click “ Cancel ” button. System discards unsaved changes. System redirects the traveler to the Trip List.	
Exception Flow		Description	
	EF-01	At Step 2, Traveler leaves mandatory fields empty. System highlights missing fields. System displays validation messages.	
	EF-02	At Step 2, Traveler fill in End Date earlier than Start Date. System validates required fields in real time Systems display error messages for invalid date.	
	EF-03	At Step 1, Traveler wants to edit non-existed trip.	
	EF-04	At Step 3, Traveler click “ Save ” button. Database failed so system displays database error.	

Business Rules	Destination is mandatory. Start Date is mandatory. End Date must be later than Start Date.
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds. Trip data shall be stored securely. System availability shall be at least 99%.
Acceptance Criteria	Given a traveler enters valid trip information When clicking Edit Then new trip information is updated And new information appears in the Trip List And a success message is displayed

Use Case ID	UC – 003		
Use Case Name	Delete Trip		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 02, 2026	Date Created	July 02, 2026
Primary Actors	Traveler		
Supporting Actors	None		
Goal	Allow traveler to delete unnecessary trip to stay organized.		
Trigger	“Delete” trip button is clicked by traveler.		
Pre-conditions	Trip exists. Traveler has permission to delete.		
Post-conditions	A deleted trip record is successfully created and new Trip List is displayed.		
Main Flow	Step	Actor Action	System Response
	1	Traveler click “Delete” trip.	System displays a confirmation notification
	2	Traveler click “Confirm” button.	System deletes trip record. System displays a success message. System redirects the traveler to the Trip List.
Alternative Flow		Description	
	AF-01	At step 2, Traveler click “Cancel” button. System discards unsaved changes. System redirects the traveler to the Trip List.	
Exception Flow		Description	
	EF-01	At Step 1, Traveler wants to delete non-existed trip.	
Business Rules	Archived trips can also be deleted. Deleting a trip permanently removes associated collaboration data.		

Special Requirements (Non-functional)	System response time shall not exceed 3 seconds. Trip data shall be stored securely. System availability shall be at least 99%.
Acceptance Criteria	Given a traveler delete unnecessary trip When clicking Delete Then the selected trip is deleted And new Trip List is updated And a success message is displayed

Use Case ID	UC – 004		
Use Case Name	Archive Trip		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 02, 2026	Date Created	July 02, 2026
Primary Actors	Traveler		
Supporting Actors	None		
Goal	Allow traveler to preserve history trip in order to be organized		
Trigger	“Archive” trip button is clicked by traveler.		
Pre-conditions	Trip exists. Traveler has permission to archive.		
Post-conditions	An archived trip record is successfully created and new Trip List is displayed.		
Main Flow	Step	Actor Action	System Response
	1	Traveler click “Archive” trip.	System displays a confirmation notification
	2	Traveler click “Confirm” button.	System archives trip record. System displays a success message. System redirects the traveler to the Trip List.
Alternative Flow		Description	
	AF-01	At step 2, Traveler click “Cancel” button. System discards unsaved changes. System redirects the traveler to the Trip List.	
Exception Flow		Description	
	EF-01	At Step 1, Traveler wants to archive non-existed trip.	
	EF-02	At Step 1, Traveler wants to archive an archived trip.	
Business Rules	Only active trips are archived. All historical information will be preserved after being archived.		
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds. Trip data shall be stored securely.		

	System availability shall be at least 99%.
Acceptance Criteria	Given a traveler archived trip When clicking Archive Then the selected trip is archived And new Trip List is updated And a success message is displayed

Use Case ID	UC – 005		
Use Case Name	Restore Trip		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 02, 2026	Date Created	July 02, 2026
Primary Actors	Traveler		
Supporting Actors	None		
Goal	Allow traveler to reactivate archived trips to continue planning in the future.		
Trigger	“Restore” trip button is clicked by traveler.		
Pre-conditions	Trip is being archived. Traveler has permission to restore.		
Post-conditions	A restored trip record is successfully created and new Trip List is displayed.		
Main Flow	Step	Actor Action	System Response
	1	Traveler click “Restore” trip.	System displays a confirmation notification
	2	Traveler click “Confirm” button.	System restores archived trip record. System displays a success message. System redirects the traveler to the Trip List.
Alternative Flow		Description	
	AF-01	At step 2, Traveler click “Cancel” button. System discards unsaved changes. System redirects the traveler to the Trip List.	
Exception Flow		Description	
	EF-01	At Step 1, Traveler wants to restore non-existed trip.	
	EF-02	At Step 1, Traveler wants to restore a restored trip.	
Business Rules	Only archived trips can be restored. All trip’s information will be restored.		
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds. Trip data shall be stored securely. System availability shall be at least 99%.		

Acceptance Criteria	Given a traveler restore archived trip When clicking Restore Then the selected trip is restored And new Trip List is updated And a success message is displayed
----------------------------	--

Use Case ID	UC – 006		
Use Case Name	Register		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 03, 2026	Date Created	July 03, 2026
Primary Actors	Traveler		
Supporting Actors	None		
Goal	Allow traveler to register an account and access personalized trip planning features.		
Trigger	"Sign up" button clicked by traveler.		
Pre-conditions	Traveler is not authenticated.		
Post-conditions	A new account is recorded successfully.		
Main Flow	Step	Actor Action	System Response
	1	Traveler clicks "Sign up"	System displays registration form
	2	Traveler enters personal information	System validates required fields in real time.
	3	Traveler clicks "Sign up" button	System validates all inputs A new account is created System displays success notification System redirects user to Trip List
Alternative Flow		Description	
Exception Flow		Description	
	EF-01	At step 2, traveler leaves empty mandatory field, system highlights missing required fields, displays validate messages.	
	EF-02	At step 2, traveler enter wrong email format or non-existed email, system highlights false fields, displays validate messages.	
	EF-03	At step 2, traveler enter existed email, system highlights false fields, displays validate messages.	
Business Rules	Each email address must be unique. Password must satisfy the password policy.		

	Each traveler can register only one account.
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds System availability shall be at least 99%.
Acceptance Criteria	Given a traveler enters valid registration information When clicking Sign Up Then a new account is created And the traveler is redirected to the Trip List And a success message is displayed.

Use Case ID	UC – 007		
Use Case Name	Log In		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 03, 2026	Date Created	July 03, 2026
Primary Actors	Traveler		
Supporting Actors	None		
Goal	Allow authenticated access for traveler to personal trips.		
Trigger	"Log In" button is clicked by traveler.		
Pre-conditions	User already has an account. User is being logged out.		
Post-conditions	Log in activity is recorded.		
Main Flow	Step	Actor Action	System Response
	1	Traveler access to TravelMate and click "Log In"	System displays profile information input form
	2	Traveler enters personal information.	System validates required fields in real time and authenticates traveler
	3	Traveler click "Log In" button	System validates all inputs System displays success notification System redirects user to Trip List
Alternative Flow		Description	
	AF-01	At step 2, Traveler forgot password so system validates traveler by verifying email, display another data input form.	
Exception Flow		Description	
	EF-01	At step 2, Traveler enters wrong username/email or wrong password, after click "Log In" so system highlights false fields, displays validate messages.	
	EF-02	At step 1, Traveler’s account is locked, inactive or the server is unavailable so system displays error messages.	
Business Rules	Only registered travelers can log in.		

	A traveler may have only one active session.
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds System availability shall be at least 99%.
Acceptance Criteria	Given a traveler enters valid registration information When clicking Log In Then log in activity is recorded And system authenticates traveler to access personal trips And the traveler is redirected to the Trip List And a success message is displayed.

Use Case ID	UC – 008		
Use Case Name	Log Out		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 03, 2026	Date Created	July 03, 2026
Primary Actors	Traveler		
Supporting Actors	None		
Goal	Allow traveler to log out to securely end my session.		
Trigger	"Sign Out" button clicked by traveler.		
Pre-conditions	User already has an account. User is being logged in.		
Post-conditions	Log out activity is recorded.		
Main Flow	Step	Actor Action	System Response
	1	Traveler access to TravelMate and click "Log Out"	System displays a confirmation notification
	2	Traveler click “Confirm”	System terminates current session. Logout activity is recorded. System redirects the traveler to the Login page. System displays a success message.
Alternative Flow		Description	
Exception Flow		Description	
	EF-01	Session already expired so traveler cannot logout.	
Business Rules	Only authenticated travelers can log out.		
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds System availability shall be at least 99%.		
Acceptance Criteria	Given a traveler already logged in When clicking Log Out Then log out activity is recorded		

	And system redirects traveler back to authentication page And a success message is displayed.
--	--

Use Case ID	UC – 009		
Use Case Name	Edit Profile		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 03, 2026	Date Created	July 03, 2026
Primary Actors	Traveler		
Supporting Actors	None		
Goal	Allow traveler to update profile so account details remain accurate.		
Trigger	"Edit Profile" button is clicked by traveler.		
Pre-conditions	User already has an account. User is being logged in.		
Post-conditions	Updated information about the account is recorded and displayed in the profile.		
Main Flow	Step	Actor Action	System Response
	1	Traveler click "Edit Profile"	System displays profile information input form
	2	Traveler enters updated personal information	System validates required fields in real time
	3	Traveler click "Save" button	System validates the input System updates profile record System displays success notification System redirects user to view profile
Alternative Flow		Description	
	AF-01	At step 3, Traveler click “Cancel” button. System discards unsaved changes. System redirects the traveler to the Profile page.	
Exception Flow		Description	
	EF-01	At step 2, traveler enter the wrong email format or non-existed email so system highlights false fields, displays validate messages.	
	EF-02	At step 2, traveler enter existed email/username so system highlights false fields, displays validate messages.	
Business Rules	Email address must remain unique. Traveler may update only their own profile.		
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds System availability shall be at least 99%.		

Acceptance Criteria	Given a traveler enter valid information When clicking Edit Profile Then new account information is updated And new information appears in view profile And a success message is displayed.
----------------------------	--

Use Case ID	UC – 010		
Use Case Name	Change Password		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 03, 2026	Date Created	July 03, 2026
Primary Actors	Traveler		
Supporting Actors	None		
Goal	Allow traveler to change password so that the account remains secure.		
Trigger	"Change password" button is clicked by traveler.		
Pre-conditions	User already has an account. User is being logged in.		
Post-conditions	Updated password is recorded.		
Main Flow	Step	Actor Action	System Response
	1	Traveler click "Change Password"	System displays password input form including old password and new password
	2	Traveler enters updated password	System validates required fields in real time
	3	Traveler click "Save" button	System verifies the old password System validates the input System updates profile record System displays success notification System redirects user to view profile
Alternative Flow		Description	
	AF-01	At step 3, Traveler click "Cancel" button. System discards unsaved changes. System redirects the traveler to the Profile.	
Exception Flow		Description	
	EF-01	Traveler enters a new password identical to the current password. System highlights false fields, displays validate messages.	

	EF-02	Traveler enters old password incorrectly. System highlights false fields, displays validate messages.
Business Rules	New password must be different from the current password. Password must comply with the password policy.	
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds System availability shall be at least 99%.	
Acceptance Criteria	Given a traveler enter valid password When clicking Change Password Then new password is updated And change password activity is recorded And a success message is displayed.	

Use Case ID	UC – 011		
Use Case Name	Share Trip		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 04, 2026	Date Created	July 04, 2026
Primary Actors	Owner		
Supporting Actors	Member, Email Service		
Goal	Allow traveler to enable collaborative trip planning.		
Trigger	“Share Trip” button clicked by owner		
Pre-conditions	User is authenticated. User is the owner of the selected trip. The selected trip exists. Collaboration is currently disabled.		
Post-conditions	Share trip activity is recorded.		
Main Flow	Step	Actor Action	System Response
	1	Owner clicks “Share Trip”	System displays a confirmation dialog
	2	Owner clicks “Confirm”	System enables Collaboration for the trip. System records activity log. System displays Member Management. System displays success message.
Alternative Flow		Description	
	AF-01	At step 2, Traveler click “Cancel” button. System discards unsaved changes. System redirects the traveler to the trip list.	
Exception Flow		Description	

	EF-01	Database unavailable. System displays an error. Collaboration is not enabled.
Business Rules	Only the Trip Owner may enable Collaboration. Collaboration can only be enabled once.	
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds. System availability shall be at least 99%. Trip data shall be stored securely.	
Acceptance Criteria	Given an authenticated traveler who is an owner When clicking share trip Then share trip feature is available And share trip activity is recorded And a success message is displayed.	

Use Case ID	UC – 012		
Use Case Name	Add Member		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 04, 2026	Date Created	July 04, 2026
Primary Actors	Owner		
Supporting Actors	Member, Email Service		
Goal	Allow traveler to add members so that more people can collaborate on the trip.		
Trigger	“Add Member” button is clicked by owner		
Pre-conditions	User is authenticated. User is the owner of the selected trip. The selected trip exists. Member has an account.		
Post-conditions	Add member activity is recorded		
Main Flow	Step	Actor Action	System Response
	1	Owner clicks “Add Member”	System displays sharing trip input form
	2	Owner entering member’s email and role	System validates required fields in real time
	3	Owner clicks “Save”	System validates member’s account System creates email sending request Email service sent share trip notification through email to member Add member activity is recorded

			System displays success notification System redirects user to member list.
Alternative Flow		Description	
	AF-01	At step 2, Traveler click “Cancel” button. System discards unsaved changes. System redirects the traveler to the trip.	
Exception Flow		Description	
	EF-01	At step 2, Traveler enter the wrong email format or non-existed email so system highlights false fields, displays validate messages.	
	EF-02	At step 2, Traveler enter existed member’s email so system highlights false fields, displays validate messages.	
	EF-03	Member limit exceed; owner can only add a certain number of members	
Business Rules	The same traveler cannot be added more than once to the same trip. New member added must be different from existed trip’s member. Member’s role cannot be owner. Member cannot add members.		
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds. System availability shall be at least 99%. Trip member data shall be stored securely.		
Acceptance Criteria	Given a traveler enter valid member’s information When clicking Add member Then a new list of the trip’s member is updated And add member activity is recorded And a success message is displayed.		

Use Case ID	UC – 013		
Use Case Name	Remove Member		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 04, 2026	Date Created	July 04, 2026
Primary Actors	Owner		
Supporting Actors	Member, Email Service		
Goal	Allow owner to remove members to manage trip participants.		
Trigger	"Remove Member" button clicked by owner		
Pre-conditions	User is authenticated. User is the owner of the selected trip. The selected trip exists. Member is already a trip's member.		
Post-conditions	Remove member activity is recorded		
Main Flow	Step	Actor Action	System Response

	1	Owner clicks “ Remove Member ” on selected member	System displays a confirmation dialog
	2	Owner clicks “ Confirm ”	System updates member list and refreshes the list Remove member activity is recorded Email service sends remove notification message to removed member System redirects the traveler to the Member list System displays a success message.
Alternative Flow		Description	
	AF-01	At step 2, Traveler click “ Cancel ” button. System discards unsaved changes. System redirects the traveler to the trip.	
Exception Flow		Description	
	EF-01	At step 1, owner remove themselves, system display error messages	
	EF-02	At step 1, owner remove member no longer exists, system display error messages	
Business Rules	Owner cannot remove themselves Member cannot remove other members Owner must remain the only Owner		
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds. System availability shall be at least 99%. Trip member data shall be stored securely.		
Acceptance Criteria	Given a traveler is an owner of the trip When clicking remove member Then a new list of the trip’s member is updated And remove member activity is recorded And a success message is displayed.		

Use Case ID	UC – 014		
Use Case Name	View Member		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 04, 2026	Date Created	July 04, 2026
Primary Actors	Owner		
Supporting Actors	Member, Email Service		

Goal	Allow owner and member to view all trip members to know who is participating.		
Trigger	“View Member” button clicked by owner/member		
Pre-conditions	User is authenticated. User is the owner of the trip or user is a trip’s member. The selected trip exists.		
Post-conditions	Member list is displayed		
Main Flow	Step	Actor Action	System Response
	1	Owner clicks “View Member”	System displays member list
Alternative Flow		Description	
Exception Flow		Description	
	EF-01	Unable to retrieve member list	
Business Rules	User has an account User is a member of the trip User can be an owner or a member		
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds. System availability shall be at least 99%. Trip member data shall be stored securely.		
Acceptance Criteria	Given a traveler is the trip’s owner or member When clicking view member Then the member list is displayed		

Use Case ID	UC – 015		
Use Case Name	Change Member Role		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 04, 2026	Date Created	July 04, 2026
Primary Actors	Owner		
Supporting Actors	Member, Email Service		
Goal	Allow owner to change member’s role to support group trip planning and give member their own selected control on the trip.		
Trigger	“Change Member Role” button clicked by owner		
Pre-conditions	User is authenticated. User is the owner of the selected trip. The selected trip exists. Member is already a trip’s member.		
Post-conditions	Change member role activity is recorded		
Main Flow	Step	Actor Action	System Response
	1	Owner selects a member. Owner selects a new role.	

		Owner clicks Save.	
	2	Owner clicks “Save”	System updates trip member role, change member role activity is recorded Email service sends change role notification message to selected member System redirects the traveler to the View Member list System displays a success message.
Alternative Flow		Description	
	AF-01	At step 2, Traveler click “Cancel” button. System discards unsaved changes. System redirects the traveler to the trip list.	
Exception Flow		Description	
	EF-01	At step 1, owner changes their role so system display error messages.	
	EF-02	At step 1, owner selects member that no longer exists so system display error messages	
	EF-03	Email service failed	
Business Rules	Owner cannot change role themselves Member cannot change role other members The new role can be the same as the old one		
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds. System availability shall be at least 99%. Trip member data shall be stored securely.		
Acceptance Criteria	Given a traveler is an owner of the trip When clicking Change member role Then a new list of the trip’s member role is updated And change member role activity is recorded And a success message is displayed.		

Use Case ID	UC – 016		
Use Case Name	Accept Invitation		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 04, 2026	Date Created	July 04, 2026
Primary Actors	Invited Traveler		
Supporting Actors	Email Service		

Goal	Allow traveler to accept a shared trip invitation to collaborate with other members.		
Trigger	“Accept Invitation” button from traveler		
Pre-conditions	User is authenticated User is not the owner of the trip User got an invitation from the trip owner		
Post-conditions	Accept invitation activity is recorded		
Main Flow	Step	Actor Action	System Response
	1	Traveler opens invitation. Traveler clicks “Accept”	System validates invitation. System adds traveler into trip. System activates assigned role. System records activity. System displays success.
Alternative Flow		Description	
Exception Flow		Description	
	EF-01	Email Service failed	
Business Rules	Only authenticated travelers can accept invitation Traveler already has an account Traveler is not the owner of the trip.		
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds. System availability shall be at least 99%. Trip member data shall be stored securely.		
Acceptance Criteria	Given a traveler is authenticated When clicking accepting invitation Then a new list of the trip’s member is updated And invitation status becomes Accept And updated list member is recorded And a success message is displayed.		

Use Case ID	UC – 017		
Use Case Name	Decline Invitation		
Created by	Ngoc Vu	Created by	Ngoc Vu
Date Created	July 04, 2026	Date Created	July 04, 2026
Primary Actors	Invited Traveler		
Supporting Actors	Email Service		
Goal	Allow user to reject a trip invitation to decline collaboration.		
Trigger	"Decline Invitation" button from traveler		
Pre-conditions	User is authenticated		

	User is not the owner of the trip User got an invitation from the trip owner		
Post-conditions	Decline invitation activity is recorded		
Main Flow	Step	Actor Action	System Response
	1	Traveler opens invitation. Traveler clicks “Decline”	System validates invitation. System marks invitation as Decline. System records activity. System displays messages.
Alternative Flow		Description	
Exception Flow		Description	
	EF-01	Email Service failed	
Business Rules	Only authenticated travelers can accept invitation Traveler already has an account Traveler is not the owner of the trip.		
Special Requirements (Non-functional)	System response time shall not exceed 3 seconds. System availability shall be at least 99%. Trip member data shall be stored securely.		
Acceptance Criteria	Given a traveler is authenticated When clicking decline invitation Then trip invitation is discarded And invitation status becomes Decline And a message is displayed.		